ShadeShift: Smart Modular Eyewear for Neurological Recovery and Visual Stress Alleviation

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***Abstract*—**Visual hypersensitivity, photophobia, and impaired light processing are common symptoms experienced by patients recovering from traumatic brain injuries (TBIs), neurological surgeries, or migraines. These conditions significantly impact the quality of life and cognitive recovery of affected individuals. Existing solutions, such as static-tint sunglasses or eye patches, fail to offer dynamic control, quadrant-based modulation, or aesthetic integration with everyday use. To address this unmet clinical and ergonomic need, we present ShadeShift, a modular, smart eyewear system that enables real-time control of light transmission through adjustable lenses embedded with electrochromic lenses and offers stroboscopic visual therapy via a Polymer Network Liquid Crystal (PNLC) film.

ShadeShift allows users to independently regulate the tint level of the lenses, thereby supporting patient vision rehabilitation, glare reduction, and stress minimization. The prototype of the system is built using a 3D-printed ergonomic frame, custom software, a low-power microcontroller, and user-operated switches, with provisions for future wireless app-based control. Our design process emphasized both functional and aesthetic considerations, ensuring that users feel comfortable wearing the device in both clinical and public settings. ShadeShift bridges the gap between neuro-assistive technology and wearable design, offering a scalable and impactful solution for a diverse user base. This paper presents the end-to-end design, implementation, testing, and future vision of ShadeShift in the context of medical wearables and personalized healthcare technologies.

Keywords*—Smart Eyewear, Neurological Recovery, PDLC, PNLC, Visual Stress, Assistive Devices, CAD, Modular Design, Stroboscopic Visual Therapy, Binasal Occlusion, TBI*

I. Introduction

Traumatic brain injuries (TBIs), post-neurosurgical recovery, and chronic neurological conditions such as migraines or post-concussion syndrome often result in visual disturbances, including photophobia, blurred vision, and hypersensitivity to ambient and dynamic lighting conditions. These impairments significantly hinder patients' ability to carry out daily activities and participate in cognitive rehabilitation. Standard light-filtering eyewear, including tinted sunglasses or manual binasal occlusion patches, fails to address the adaptive needs of patients who require specific, situational control over their visual reception.

The need for personalized, electronically modulated visual aids has led to the development of ShadeShift. This innovative eyewear concept was born from clinical insights and feedback from neurological rehabilitation specialists and patients alike. Unlike traditional eyewear, ShadeShift offers programmable light modulation using electrochromic lenses embedded in each lens. The idea of ShadeShift aligns with recent trends in medical wearable technology, where personalization, modularity, and non-invasiveness are prioritized. The glasses are engineered with ergonomic considerations to ensure long-term wearability, embedded microelectronics for tint control, and are manufactured using rapid prototyping techniques such as PLA 3D printing. This paper elaborates on the design, development, and testing of ShadeShift and evaluates its potential as a clinical and commercial solution for visual stress management.

II. Background

The visual system is a highly complex neurological subsystem that is particularly vulnerable following brain trauma or dysfunction. In clinical rehabilitation, light sensitivity is not merely a comfort issue—it has been shown to interfere with oculomotor training, sensory integration, and neuroplasticity-driven recovery. Visual impairments such as peripheral overload, delayed adaptation to lighting changes, and specific field disruptions require targeted management strategies [1]. Wavelengths of light, particularly blue-green light, which is known to trigger migraines and photophobia, can also be detrimental to patients, as they would need to have different filters for colors at all times [2].

Standard interventions, including binasal occlusion therapy or prism lenses, address some of these issues, but they are generally static, manually applied, and lack dynamic response capabilities. Moreover, patients undergoing neurorehabilitation often require devices that support stroboscopic training, oculomotor isolation, or field-specific visual filtering, depending on their therapy regimen [3].

In parallel, developments in electro-optic materials—especially PDLC and PNLC technologies—have enabled precise control over light transmittance. These materials change their optical properties in response to electric fields and have been successfully deployed in environments requiring rapid, reversible dimming. Some research has been done to further the usage of these films in eyewear applications [4].

Understanding this interplay between neurovascular therapy needs and responsive optics is essential for the development of next-generation assistive eyewear that goes beyond passive tinting and supports therapy-driven customization.

III. Related Work

Vision therapy has increasingly integrated technological aids to support neurological recovery, particularly for patients with traumatic brain injuries (TBIs), post-concussive syndrome, and other visual processing disorders. Commercial devices such as the Senaptec Strobe glasses use liquid crystal shutters to intermittently obstruct vision, enhancing visual-motor coordination and reaction time through stroboscopic training [5]. While effective in sports training and rehabilitation settings, these glasses lack quadrant-specific control and are not optimized for everyday use or long-duration wear.

Academic research has explored various aspects of smart eyewear. For example, a proposed prototype integrates a form of PDLC/PNLC called PLATO (Portable Liquid-crystal Apparatus for Tachistoscopy via visual Occlusion) [4]. The paper details its design objectives—comfort, rapid switching (around 1 ms on, effectively <4 ms off), adequate information extinction, safety, and reliability—and discusses its operating characteristics. Potential applications are summarized, encompassing research in areas like automobile driving behavior and visual-motor coordination, clinical ophthalmology (e.g., VEP testing), and use as a high-transmission viewing device for field-sequential stereoscopic displays. Similarly, there are glasses currently available by Ampere, which are the Dusk Smart Sunglasses that feature electrochromic lenses, allowing users to manually adjust tint levels for different lighting conditions. [7] However, they do not support graded occlusion or therapeutic functionalities essential for PCS management.

In clinical environments, manual occlusion therapy remains a standard approach to managing visual field deficits or binocular dysfunction. These include patching or using opaque films to restrict vision, particularly for pediatric and post-injury patients. However, these tools offer no dynamic adjustment and often suffer from low user compliance due to discomfort or aesthetic limitations. There is a paper that investigates the use of binasal occlusion (BNO) in neurorehabilitation, where selective blocking of peripheral vision helps patients with visual neglect or processing deficits [8]. The study employed BNO filters that reduced visual overload and improved spatial awareness in individuals with brain injuries. The findings support the efficacy of binasal occlusion as a non-invasive method to enhance visual anchoring and mitigate sensory disturbances, validating its inclusion in therapeutic eyewear.

While each of these works has contributed to the understanding and development of visual modulation tools, there remains a clear gap in the availability of lightweight, customizable, and tint-specific visual aids designed specifically for continuous therapeutic wear. This gap underscores the need for new solutions that integrate dynamic optical control with ergonomic and modular design principles for real-world clinical applications.

IV. Design and Implementation

The custom eyewear frame we designed is composed of several functional regions: nose bridge, dual lens frames, temple arms, and frame base. Each of the lenses includes a compartment into which the electrochromic lenses are embedded. The frames were designed using Tinkercad and were printed using PLA filament to rapidly print and test our prototype (see Fig. 1). The frame design includes wells for magnets to be embedded, allowing filters and shielding units to be added or replaced as needed.

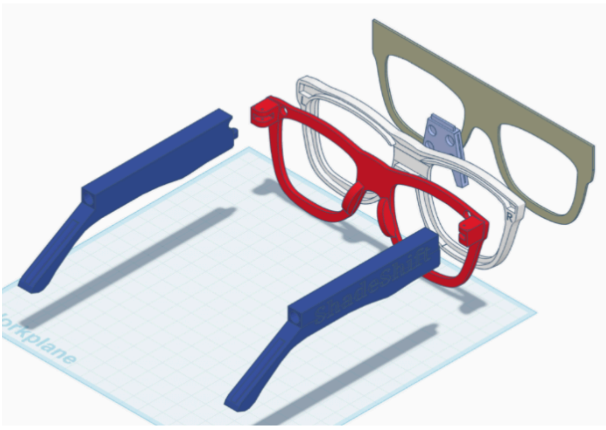

Fig 1. Exploded view of the ShadeShift eyewear.

The ShadeShift system comprises two primary components: the modular smart eyewear and an external control unit housing the electronics. The eyewear frame features interchangeable magnetic lenses in multiple hues, enabling users to tailor light filtration based on comfort or therapy needs. Adjustable binasal occlusion strips are integrated with the magnetic clips to aid visual focus and reduce sensory overload. The frame is 3D-printed for lightweight wearability and quick customization.

The external enclosure—also custom-designed—is built to contain the rechargeable battery, microcontroller, and custom circuitry responsible for controlling the tint levels and strobe functionality (see Fig. 2). This compact unit interfaces with the eyewear via a wired tether, providing real-time control of the electrochromic lens tint transitions and stroboscopic training intensity of the PDLC. Both components were iteratively tested and refined to optimize usability, durability, and effectiveness.

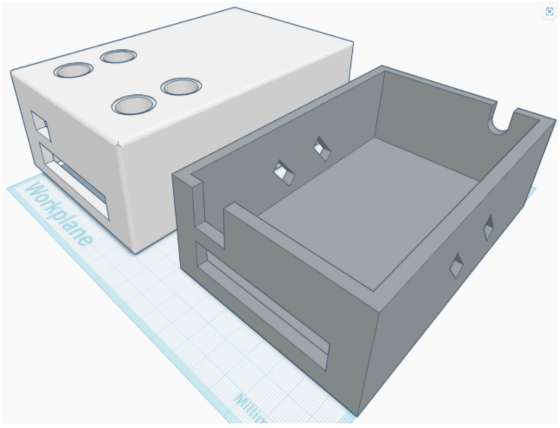

Fig. 2 The external enclosure, which will house the battery, electronics, and the buttons for control.

V. Prototyping and Testing

Our initial prototype used the components of an electrochromic eyeglass where in which the lenses were harvested for the final prototype. The lenses were controlled by an ESP32 Dev Board and were breadboarded to have buttons change the tint level of the lenses via Pulse Width Modulation (PWM) to limit the voltage being applied to the lenses. When limiting the voltage, it allows for the lenses to darken with more voltage and become transparent when no voltage is applied. See Figure. 3.

Fig. 3 A demonstration of the electrochromic lenses changing via the ESP32.

We also tested functional components like the stroboscopic training using an external controller powered by the same battery used for the ESP32. During our research, we found that the PDLC film was considered for other eyewear projects and that there was a paper that detailed how the film was used for their need [9]. This paper proved helpful when trying to integrate the film into our design and the challenges of using the film for our use case.

Real-time tests verified fast transition times of the PDLC. The magnetic lens stability and user comfort were also tested once fully assembled. The hands-on video demonstration and real-world assembly confirmed ease of attachment, system responsiveness, and the practicality of user adjustments.

Our completed prototype was 3D-printed using lightweight, durable filament to test the ergonomics and assembly of the modular ShadeShift eyewear. The design included magnetic clip-on lenses, engraved side arms for branding, and support for integrated binasal occlusion strips. See Figures 4 and 5.

Fig. 4 The 3D printed parts, shown are the temples and the frame for the PDLC film.

Fig. 5 A 3D CAD drawing of the front of the frames.

This prototyping phase helped validate the Shade Shift system's modularity, adaptability, and therapeutic promise in real environments.

VI. Results

The final Shade Shift prototype successfully demonstrated all intended functionalities. The eyewear supported interchangeable color lens attachments, allowing users to switch between green, blue, and rose-tinted filters for customized visual comfort. The binasal attachment was also able to be magnetically attached and could be used in tandem with the color lens attachments. The frames featured integrated PDLC film, enabling dynamic switching between transparent and opaque states to facilitate the stroboscopic visual training. Finally, the electrochromic lenses allowed for real-time dimming of the lens tint. See Figure 6.

Fig. 6 The fully assembled glasses with the electrochromic lenses, binasal and color clip-on attachments, and the PDLC stroboscopic attachment.

The electrochromic lenses offered adjustable tint levels to adapt to changing light conditions, while the binasal occlusion strips effectively narrowed peripheral vision, aiding focus and sensory processing. Testing confirmed smooth transitions, strong magnetic attachment, and reliable electronics performance. The results validate Shade Shift as a versatile and therapeutic eyewear solution for individuals recovering from neurological trauma [10].

VII. Challenges and Limitations

Design Complexity: One of the most significant design challenges was achieving a frame structure that could simultaneously accommodate segmented PNLC films, embedded wiring, and ergonomic fit for the user. Unlike conventional eyewear, our design required internal routing channels for electrical connections to each lens, without compromising comfort or aesthetics. Integrating these components while maintaining precise optical alignment proved difficult. Multiple CAD iterations and physical prototypes were needed to refine the lens housings, interlocking features, and wire exit points. These iterations aimed to optimize both manufacturability and user experience, balancing structural integrity with lightweight form.

Tint Adjustment Speed: The responsiveness of the PNLC films varied depending on ambient temperature and applied voltage levels. Under certain lighting or thermal conditions, the switching speed between transparent and opaque states exhibited latency, which limited the effectiveness of real-time visual modulation. In a therapeutic context, especially for stroboscopic training or fast dynamic occlusion, this delay could reduce treatment efficacy or user satisfaction. Although within acceptable limits for general dimming, the inconsistency highlighted the need for improved or alternative materials in future iterations to ensure reliable performance across environments.

Weight and Component Placement: The temple arms of the eyewear, while ergonomic, did not offer sufficient space to embed all electronic components, including the microcontroller, PCB, and rechargeable battery. To maintain the device's low-profile and modular design, these components were housed in an external enclosure tethered to the eyewear. While this solution enabled full functionality, it introduced drawbacks in terms of portability and cable management. Users may find the external unit cumbersome for long-term wear, particularly in mobile or outdoor scenarios. This limitation has informed our goal of miniaturizing the control circuitry and exploring integrated battery solutions in future designs.

VIII. Future Work

Future iterations of the ShadeShift system will focus on improving integration, responsiveness, and user experience to enhance its clinical utility and wearability. One of the primary objectives is to eliminate the reliance on an external electronics enclosure by creating a miniaturized printed circuit board (PCB) and embedding it directly within the temple arms of the eyewear. This would be accompanied by the integration of a smaller, lightweight battery that supports extended usage while maintaining overall ergonomics and portability. Achieving this level of hardware integration will significantly improve user comfort and device portability, making the system more practical for daily therapeutic use.

Another area of ongoing development is the thermal response of the PDLC/PNLC films used for dynamic light modulation. The current films exhibit variations in switching speed and consistency depending on ambient temperature. To address this, we plan to investigate the application of thermal insulation techniques and the use of alternative or hybrid electrochromic materials that offer faster, more reliable transitions across a wider range.

User interaction will also be enhanced through the incorporation of capacitive touch interfaces embedded into the frame. These touch-sensitive controls would allow users to switch lens tint levels and strobe speed seamlessly, eliminating the need for mechanical switches or external devices, and improving the intuitiveness and accessibility of the system.

To assess long-term reliability and broad usability, further testing will evaluate the performance of the PDLC/PNLC films under diverse ambient conditions and among varied user demographics, including patients with different neurological profiles and light sensitivity thresholds. These tests will inform adjustments in film calibration and system responsiveness.

Finally, we aim to conduct structured usability trials in collaboration with neurologists, optometrists, and physical therapists. These clinical evaluations will provide essential feedback on comfort, therapeutic efficacy, and overall adoption potential in real-world rehabilitative settings. Insights from these trials will guide iterative refinement and help validate Shade Shift's potential as a scalable medical wearable device.

To put into concise points,

1. Manufacture a miniaturized PCB and embed it inside the temple arms.
2. Integrate a smaller battery to eliminate the external control box.
3. Optimize film switching speed with insulation or hybrid materials.
4. Add capacitive touch sensors for easier control and accessibility.
5. Test film performance across varied users and conditions.

6. Conduct clinical trials with medical specialists for usability evaluation.

IX. Conclusion

ShadeShift presents a novel integration of tint-specific optical modulation, ergonomic design, and modular architecture for assistive eyewear applications. Developed through iterative prototyping and engineering validation, the device addresses a significant clinical gap by offering customizable control for individuals recovering from neurological trauma. The use of PDLC/PNLC film technology enables dynamic, localized stroboscopic visual training. The electrochromic lenses offer tint adjustment, eliminating the need for static sunglasses. The magnetic clip on binasal occlusion attachments, along with the magnetic color-changing frames, distinguish ShadeShift from conventional, static-tint, and color hue solutions.

The system emphasizes both functionality and usability, incorporating features such as magnetic lens attachments, external control modules, and a lightweight, 3D-printed frame. These design choices were guided by feedback from real-world use cases and user comfort considerations, supporting extended wear in therapeutic settings.

Initial testing confirmed successful implementation of key functions, including real-time tint modulation and modular component integration. Ongoing work will focus on embedding electronics within the frame, optimizing film responsiveness, and conducting clinical evaluations to assess long-term usability and therapeutic benefit. With continued refinement, ShadeShift has the potential to evolve from a research prototype into a clinically deployable, commercially viable solution for patients experiencing visual stress and photophobia.

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